

Maciej Tkacz

Robotics Engineer - Hardware, Control Software and Rapid Prototyping

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PROFILE

I build machines end to end and write the software that runs them. As a personal project I designed and built a SCARA robotic arm from scratch: mechanics in CAD, 3D-printed structure, NEMA stepper drives with TMC/DRV drivers, microcontroller electronics, and Python/C++ control software with forward and inverse kinematics. I now do the same commercially as a Robotics Engineer at ASTOR, where I program Kawasaki and Epson industrial robots, build digital twins of production lines, and design, build and commission demonstration stations - machines whose entire job is to run reliably in front of an audience, without a technician standing next to them. Mechanical Engineering student at Cracow University of Technology and certified Software Technician, so the mechanics, the electronics and the code come from the same person.

WHAT I BRING

- Whole machines, not parts: mechanical concept and CAD, 3D-printed structure, actuator and driver selection, microcontroller electronics, wiring, assembly and bring-up. One complete robot built alone, plus robot cells built and commissioned professionally.
- Control software on real hardware: Python and C++ for robot motion and hardware interaction, forward and inverse kinematics verified against the physical arm, Arduino/C for low-level control, Python scripting for automation. ROS from personal projects, self-taught rather than commercial.
- Industrial robots: Kawasaki and Epson - certified courses, teach pendant, motion programs, complete application cycles, cell testing and commissioning.
- CAD and additive manufacturing: Fusion 360, SolidWorks, Autodesk Inventor, AutoCAD, plus FDM, SLA and MJF run in a production shop with process parameters, post-processing and quality control. I know the difference between a part that prints and a part that prints the same way a hundred times.
- Simulating before building: digital twins of production lines and robot cells - reach, motion sequences, collisions and cycle behaviour validated before anything is cut, printed or wired. A cheap way to kill a bad iteration early.
- How I work: I take a problem across mechanics, electrics and software until it actually runs, and I would rather build the next iteration than write a report about the last one.

SELECTED PROJECT

SCARA Robotic Arm - designed, built and programmed from scratch | Personal R&D project

- Took the arm from a blank page to a working machine: mechanical concept, CAD, kinematic layout and gear ratios, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Wrote the control software in Python and C++, implementing forward and inverse kinematics and testing it against the physical arm rather than only on paper.
- Designed every printed part for stiffness and assembly in PET-G and carbon-fibre reinforced material, iterating through repeated print, assemble and test cycles.
- Debugged the machine as a whole - the point where mechanics, electronics and software fail each other is the part no course prepares you for, and the part I enjoy most.

EXPERIENCE

Robotics Engineer | ASTOR, Krakow, Poland | Jul 2026 - present

- Design, build and commission demonstration stations presenting Kawasaki and Epson robots in different applications - mechanical assembly, hardware-software integration, programming and bring-up. These machines have to work in front of customers, repeatedly, on demand.
- Program Kawasaki and Epson industrial robots: motion programs, teach pendant work and complete application cycles.
- Build digital twins of production lines and robot simulations, validating reach, motion sequences and process behaviour before anything is built physically.
- Test and troubleshoot robot cells end to end, across mechanics, electrics and robot software.

Application Engineer | AIAutomation, Poland | Jan 2025 - May 2026

- Built robotics simulations and digital twins of production workcells in Visual Components for automotive clients.
- Wrote Python scripts to automate simulation logic, optimize processes and improve simulation performance.
- Developed robot paths, motion sequences, process logic, collision checking and cycle flows, and optimized workcell layouts and 3D geometry.
- Worked from customer documentation and engineering standards, and presented assumptions and improvement proposals directly to client engineers.

3D Printing and CAD Design Specialist | Cubic Inch Additive Manufacturing, Piaseczno, Poland | Jun 2023 - Aug 2023

- Ran FDM, MJF and SLA printers in a production environment: process parameters, post-processing and quality control.

- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing.
- Supported the introduction of a new SLA technology, documenting tests and technical observations.
- Coordinated production tasks in a 10-person team, balancing quality, manufacturability and deadlines.

Robotics and 3D Printing Intern | ASTOR Robotics Center, Krakow, Poland | May 2022

- Assembled mechanical hardware and 3D-printed components for Kawasaki robots and Astorino robot platforms.
- Programmed Kawasaki robots using teach pendant workflows and wrote basic motion programs.
- Tested robot and workstation operation.

EDUCATION

- Cracow University of Technology - Mechanical Engineering, Engineer's degree in progress (Oct 2024 - present). Coursework: dynamic systems modelling, automation and control, analytical mechanics, mechatronics.
- PKMechPower Student Research Group, Mechanical Section - CAD and 3D-printed parts for a student vehicle, including driver seat and brake-system components, optimized for mass and strength.
- Zespol Szkol Informatycznych, Kielce - Software Technician diploma (Sep 2019 - Apr 2024).

CERTIFICATES

- Kawasaki robot operation and programming - integrator course with certificate, ASTOR Robotics Center.
- Epson industrial robot programming - certified course.
- Python programming courses, plus self-directed Python and C++ development in robotics and automation projects.

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